

Gödel’s Ontological Argument in Lean 4: the 1970 inconsistency, modal collapse, and the emendations

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Abstract

We give a Lean 4 formalization of Gödel’s ontological argument and of the three emendations of it, with a machine-checked record of exactly what each result depends on.

Gödel’s own 1970 axioms are *inconsistent*. This was discovered in 2013 by the theorem prover Leo-II and analysed by Benzmüller and Woltzenlogel Paleo [2]; the mathematics of Section 3 is theirs. The culprit is a single missing conjunct: Gödel’s definition of *essence* does not require that x actually have φ , so the empty property is an essence of every individual, and Axiom 5 then demands that it be necessarily instantiated. The refutation needs Axioms 1, 2 and 5 only, and *no frame condition whatsoever*: it lives in K .

Scott’s 1987 variant restores the conjunct. We formalize the argument Gödel intended — Theorems 2 and 3, concluding $\Box\exists x G(x)$ — and find that it needs only *symmetry* of the accessibility relation, the modal logic B , rather than the $S5$ in which it is usually presented. We then formalize the price: *modal collapse*, $p \rightarrow \Box p$, so that nothing is contingent. Read as a condition on the frame rather than on sentences, collapse forces the accessibility relation to be *equality* — every world sees itself and nothing else — so $\Box$ and $\Diamond$ become the identity operation and no modal content survives. That settles the corresponding question about models: Scott’s axioms are satisfiable over a world set of any size whatsoever, but only as a disjoint heap of isolated one-world models. There is no non-degenerate model to be found.

All of this presupposes symmetry, and we show it cannot be dispensed with. An explicit model over a reflexive transitive frame — an $S4$ frame — satisfies Scott’s axioms while refuting Theorem 3, modal collapse and the equality of the frame. So Scott’s axioms do not entail modal collapse by themselves; they entail it in B and $S5$, which is where the argument is always presented. The price is better read as a price of the frame strength the argument needs than of the axioms alone.

We then formalize the three emendations that repair this. Anderson’s drops the implausible half of Axiom 1 and strengthens God-likeness and essence to biconditionals, which makes uniqueness especially direct — though uniqueness is available in Scott’s system too, from A1b alone. Fitting’s makes positivity a predicate on *extensions* rather than intensional properties; its Theorem 3 needs no frame condition at all, and the reason connects directly to the previous paragraph. Hájek’s merges two axioms into one weaker one, and the resulting definition of God-likeness makes his Axioms 4 and 5 superfluous: Theorem 3 follows from two of his four axioms. All three avoid modal collapse, and we exhibit two-world $S5$ models certifying it.

For Fitting’s variant we also separate the de re and de dicto readings of the conclusion. Both are theorems on the full axiom set; on his Part I axioms, before A4, they come apart, and we give the two-world model that separates them.

Finally, one small structure does double duty. A model with one world and one individual satisfies Axioms 1–4 but refutes Gödel’s Axiom 5, certifying that the inconsistency

is not an artefact of our encoding; and it satisfies *all* of Scott’s axioms, certifying that his system is consistent. The contrast between those two facts — same model, same notion of positivity, opposite verdicts on Axiom 5 — is the sharpest statement of what the missing conjunct costs.

1 Introduction

Gödel’s ontological argument is a derivation, in second-order modal logic, of the conclusion that necessarily there exists a God-like being. It descends from Leibniz and ultimately from Anselm, but Gödel’s version is unusually sharp: a primitive predicate P (“is a positive property”), three definitions, five axioms, three theorems. Gödel discussed the manuscript with Dana Scott, who circulated a slightly different version; it is Scott’s variant that most of the subsequent literature analyses, and the two differ in exactly one conjunct in one definition.

The difference is not cosmetic. In 2013 Benz Müller and Woltzenlogel Paleo ran the routine consistency check that precedes any serious attempt at machine proof, and the higher-order prover Leo-II reported that Gödel’s axioms were inconsistent [2]. Everything follows from them, including $\Box \exists x G(x)$; the 1970 argument therefore establishes nothing whatever. Gödel’s omission had been remarked on before — Hazen called it an oversight — but its consequence had gone unnoticed for four decades.

Scott’s version is another matter. It is consistent, the argument goes through, and the objection that has dogged it since Sobel is not unsoundness but *modal collapse*: the axioms entail that every truth is a necessary truth.

This note formalizes all five systems in Lean 4, in 378 lines of code with no library dependencies. The point of doing so is not doubt about the mathematics, which is short and entirely comprehensible, but bookkeeping: in a formalization every axiom and every frame condition a result consumes appears as a named hypothesis of that result, and Lean’s `#print axioms` makes the foundational dependencies visible too. Several statements below consequently carry their hypotheses more precisely than the prose versions do.

What is and is not new here. None of the mathematics. The inconsistency, the Empty Essence Lemma and the refutation of Section 3 are due to Benz Müller and Woltzenlogel Paleo [2], whose §4.1 the derivation follows; Theorems 1–3 are Gödel’s; modal collapse is Sobel’s [18]; the emendations are Anderson’s [1], Fitting’s [9] and Hájek’s [13], and all three were mechanised in Isabelle/HOL by [11, 3, 5], from which the definitions used here are taken. The frame reading of modal collapse (Theorem 17) is standard, and the superfluosity of Hájek’s A4 and A5 (Theorem 30) is reported in [5].

What this note offers is an independent mechanization in a different foundation, with a different discipline: axioms and frame conditions are hypotheses of the theorems that use them rather than global postulates, so the accounting in Section 9 is carried in the statements and checked; and the models are explicit, constructive proof terms rather than the output of a model finder. Its purpose is corroboration and exposition, not discovery — unlike the work it follows, nothing here was found automatically. Readers wanting the research frontier should go to the literature cited above, and to the ongoing dispute over the ultrafilter reading [16, 6].

The small accounting results of Sections 9, 7 and 8.6 — that the 1970 refutation needs no frame condition, that symmetry is not weakenable towards $S4$, that Scott’s axioms admit models at every cardinality, and the location of the *de re/de dicto* separation — are ones we

have not found stated, but they are elementary consequences of results in the cited work, and we do not claim them as new. Wernhard [19] computes a frame condition for Scott’s Theorem 3 by second-order quantifier elimination and leaves open whether it is the weakest; that question is the nearest thing to an opening this material still has.

2 The system

2.1 Semantics

We work with a shallow embedding of quantified modal logic, which is the standard device for this kind of mechanization and is what makes the formalization short.

Fix a type W of possible worlds and a type I of individuals. A *sentence* is a map $W \rightarrow \text{Prop}$, and a *property* is a map $I \rightarrow W \rightarrow \text{Prop}$; a property thus has an extension that may vary from world to world. Let $r \subseteq W \times W$ be an accessibility relation. We write

$$(\Box_r p)(w) := \forall v (r w v \rightarrow p(v)), \quad (\Diamond_r p)(w) := \exists v (r w v \wedge p(v)).$$

Quantifiers over individuals and over properties are read world by world, with a constant domain. Quantification over properties ranges over *all* maps $I \rightarrow W \rightarrow \text{Prop}$; see Section 11 for why this matters.

No condition is imposed on r as part of the semantics. Where a derivation needs one it is stated as a hypothesis of that derivation and nowhere else. This is the whole reason for keeping r explicit rather than fixing it to the universal relation, and Section 9 collects what it buys.

An axiom is asserted to hold *at every world*. This is the ordinary reading, and Section 3 uses it in an essential way.

2.2 Definitions

The sole nonlogical primitive is $P : (I \rightarrow W \rightarrow \text{Prop}) \rightarrow (W \rightarrow \text{Prop})$, read “ φ is a positive property”, and itself world-dependent: Axiom 4 asserts $P(\varphi) \rightarrow \Box P(\varphi)$, which would be idle if P were not allowed to vary.

Definition 1 (God-likeness). $G(x) := \forall \varphi (P(\varphi) \rightarrow \varphi(x))$.

Definition 2 (Essence). Gödel’s 1970 definition and Scott’s 1987 emendation are

$$\begin{aligned} \varphi \text{ ess } x &:= \forall \psi \left(\psi(x) \rightarrow \Box \forall y (\varphi(y) \rightarrow \psi(y)) \right), \\ \varphi \text{ ess}_S x &:= \varphi(x) \wedge \forall \psi \left(\psi(x) \rightarrow \Box \forall y (\varphi(y) \rightarrow \psi(y)) \right). \end{aligned}$$

Both read: everything x has, φ necessarily carries with it. Scott adds the requirement that x actually *have* φ . That conjunct is the entire difference between the two systems, and everything below turns on it.

Definition 3 (Necessary existence). $\text{NE}(x) := \forall \varphi (\varphi \text{ ess } x \rightarrow \Box \exists y \varphi(y))$, and $\text{NE}_S(x)$ is the same with ess_S in place of ess .

2.3 The axioms

A1a. $P(\neg\varphi) \rightarrow \neg P(\varphi)$.

A1b. $\neg P(\varphi) \rightarrow P(\neg\varphi)$.

A2. $P(\varphi) \wedge \Box \forall x (\varphi(x) \rightarrow \psi(x)) \rightarrow P(\psi)$.

A3. $P(G)$.

A4. $P(\varphi) \rightarrow \Box P(\varphi)$.

A5. $P(NE)$ (Gödel) or $P(NE_S)$ (Scott).

A1 is stated by both authors as the biconditional; we split it because the two halves are used by different results. *Gödel's system* is A1a, A1b, A2, A3, A4 together with $P(NE)$; *Scott's system* is the same with $P(NE_S)$.

3 Gödel's 1970 axioms are inconsistent

The derivation has two moving parts. The first is Gödel's own Theorem 1, and is uncontroversial; it is shared by both systems.

Lemma 4 (Theorem 1). *Assume A1a and A2. If $P(\varphi)$ holds at w , then $(\Diamond_r \exists x \varphi(x))(w)$.*

Proof. Suppose not, so that φ has empty extension at every world accessible from w . Let ψ be arbitrary. Then $\Box_r \forall x (\varphi(x) \rightarrow \psi(x))$ holds at w vacuously: at any accessible v and any y , the antecedent $\varphi(y)$ fails. By A2 together with $P(\varphi)$ we get $P(\psi)$ at w . As ψ was arbitrary, every property is positive at w ; in particular both $P(\neg\varphi)$ and $P(\varphi)$, contradicting A1a. $\square$

The second part is where the 1970 system comes apart.

Lemma 5 (Empty Essence Lemma [2]). *Let e be the property with empty extension at every world. Then for every individual x and every world w , $e \text{ ess } x$ holds at w .*

Proof. Let ψ be any property with $\psi(x)$ at w ; we must show $\Box_r \forall y (e(y) \rightarrow \psi(y))$ at w . Let v be accessible from w and y any individual. The antecedent $e(y)$ is false at v by construction, so the implication holds. $\square$

No axiom, no frame condition and no classical reasoning is used here: the statement is a triviality about vacuous implication. That is precisely what makes it dangerous. Definition 2 places a *demand* on φ — that it necessarily entail everything x has — and a property that nothing satisfies meets every such demand for free.

Remark 6. Benz Müller and Woltzenlogel Paleo observe that some philosophers may prefer to run the argument with the property of *self-difference*, $\lambda x. x \neq x$, in place of the empty property. Under full extensionality the two have the same denotation and the derivation is indifferent to which is used.

Theorem 7 ([2]). *Suppose W is nonempty. Then A1a, A2 and $P(NE)$ are jointly unsatisfiable, over an arbitrary accessibility relation r .*

Proof. By A5, $P(NE)$ holds at every world, so Lemma 4 gives

$$\text{for every } w \in W \text{ there are } v \text{ with } r w v \text{ and } x \in I \text{ with } NE(x) \text{ at } v. \quad (*)$$

Pick any $w_0 \in W$ and apply $(*)$, obtaining such a v and x . Unfolding, $NE(x)$ at v says $\forall \varphi (\varphi \text{ ess } x \rightarrow \Box_r \exists y \varphi(y))$. Instantiate at $\varphi := e$. By Lemma 5 the antecedent holds, so $\Box_r \exists y e(y)$ holds at v : at every world accessible from v , something has the empty property. Nothing has the empty property at any world. Hence

$$\text{no world is accessible from } v. \quad (\dagger)$$

But $(*)$ holds at *every* world, v included, because the axioms do. Applying it at v produces a world accessible from v , in direct contradiction with $(\dagger)$. $\square$

Note that neither A3 nor A4 is used: the argument never needs a God-like being to exist, because Lemma 4 is applied to NE directly. The last step is what keeps the frame conditions at zero. A cruder argument would cash $\Box_r \exists y e(y)$ into a witness by assuming r serial, at the cost of working in D ; that is unnecessary, since the axioms are asserted everywhere and Lemma 4 can simply be reapplied at v .

Proposition 8. *For every individual x and every world w , $e \text{ ess}_S x$ fails at w .*

Proof. It would require $e(x)$ at w , and e has empty extension. $\square$

So the Empty Essence Lemma — the only input to Theorem 7 beyond Theorem 1 — is unavailable in Scott's system, and the derivation stops. This does not by itself show Scott's system consistent; that is Proposition 21 below.

4 Scott's variant: the positive argument

We now carry out the argument Gödel intended. Throughout this section the axioms are Scott's.

Lemma 9. *Assume A1b. If $G(x)$ and $\psi(x)$ both hold at w , then $P(\psi)$ holds at w .*

Proof. If $P(\psi)$ failed at w then $P(\neg\psi)$ would hold there by A1b; since x is God-like it has every positive property, so $\neg\psi(x)$ at w , contradicting $\psi(x)$. $\square$

Theorem 10 (Theorem 2). *Assume A1b and A4. If $G(x)$ holds at w , then $G \text{ ess}_S x$ holds at w . No frame condition is needed.*

Proof. The first conjunct of ess_S is $G(x)$ at w , which is the hypothesis. For the second, let ψ satisfy $\psi(x)$ at w . By Lemma 9, $P(\psi)$ at w , so by A4, $P(\psi)$ holds at every world v accessible from w . Let v be such a world and let y satisfy $G(y)$ at v . Being God-like, y has every property positive at v , and ψ is one; so $\psi(y)$ at v . This is exactly $\Box_r \forall y (G(y) \rightarrow \psi(y))$ at w . $\square$

Lemma 11. *Assume A1b, A4 and $P(NE_S)$. If $G(g)$ holds at w , then $\Box_r \exists y G(y)$ holds at w .*

Proof. $P(NE_S)$ at w and $G(g)$ at w give $NE_S(g)$ at w . By Theorem 10, $G \text{ ess}_S g$ at w . Instantiate $NE_S(g)$ at $\varphi := G$. $\square$

This is the step Scott's conjunct rescues. In Gödel's system the same move, made with $\varphi := e$, is what produced Theorem 7.

Lemma 12. *Assume Scott's axioms and that r is symmetric. Then a God-like individual exists at every world.*

Proof. Let w be any world. By A3 and Lemma 4 there are v with $r w v$ and g with $G(g)$ at v . By Lemma 11, $\Box_r \exists y G(y)$ holds at v . By symmetry $r v w$, so $\exists y G(y)$ holds at w . $\square$

Theorem 13 (Theorem 3). *Assume Scott's axioms and that r is symmetric. Then $\Box_r \exists x G(x)$ holds at every world.*

Proof. Combine Lemma 12 with Lemma 11. $\square$

This is the conclusion of the ontological argument, and it is worth being plain about what it is. It is a *conditional*: it says that granting Scott's five axioms commits one to $\Box \exists x G(x)$. Whether they should be granted is not a question a proof assistant can address. What a proof assistant *can* do is compute the rest of the bill, which is the subject of the next section.

5 The price: modal collapse

Theorem 14 (Modal collapse, Sobel). *Assume Scott's axioms and that r is symmetric. Then for every sentence p and every world w , if p holds at w then $\Box_r p$ holds at w .*

Proof. By Lemma 12 there is g with $G(g)$ at w , and by Theorem 10, $G \text{ ess}_S g$ at w . Regard p as a property, $\psi_p := \lambda y. p$, constant in the individual. Then $\psi_p(g)$ at w is just p at w , which holds; so the second conjunct of ess_S gives

$$\Box_r \forall y (G(y) \rightarrow p) \quad \text{at } w.$$

By Theorem 13, $\Box_r \exists x G(x)$ at w . So at each world accessible from w there is a God-like individual, and hence p holds there. $\square$

Corollary 15. *If r is in addition reflexive, then $p \leftrightarrow \Box_r p$ for every sentence p : the distinction between truth and necessary truth disappears.*

Nothing is contingent. This is Sobel's objection, and it is worth stressing that it applies to Scott's *consistent* system: it is not a symptom of the 1970 defect, and repairing that defect does not touch it. Under the *S5* reading the consequence can be put more starkly still.

Theorem 16. *Assume Scott's axioms with r the universal relation. Then any two worlds are equal: W has at most one element.*

Proof. Apply Theorem 14 to the sentence $p := \lambda u. u = w$, which holds at w . Then $\Box_r p$ at w , and since every world is accessible, $v = w$ for every v . $\square$

5.1 Collapse as a condition on the frame

The same instance of Theorem 14 says something more general, and more informative, once r is not assumed universal. Modal collapse is usually stated as a fact about sentences; its semantic content is a fact about the accessibility relation.

Theorem 17. *Assume Scott's axioms and that r is symmetric. Then r is equality: $r w v$ holds if and only if $v = w$.*

Proof. For one direction, apply Theorem 14 at w to $p := \lambda u. u = w$, which holds at w ; this gives $\Box_r p$ at w , that is, $r w v \rightarrow v = w$. For the other, Lemma 4 applied to A3 produces some v with $r w v$; by what we have just shown $v = w$, so $r w w$. $\square$

Two features of this deserve comment. The frame conditions are *derived*, not assumed: reflexivity and seriality are consequences of the axioms, not hypotheses added to make the argument work. And the conclusion is not that there are few worlds, but that no world can see another.

Corollary 18. *Under the same hypotheses, $\Box_r p \leftrightarrow p$ and $\Diamond_r p \leftrightarrow p$ for every sentence p .*

So the modal operators are the identity operation. Theorem 16 is the special case in which r is additionally required to be universal: equality and the universal relation agree only on a one-point set. But the general statement is the one that bites. The argument is conducted in a modal language about what is possible and what is necessary, and its own premises leave that language with nothing to say: every world is an isolated point, $\Box$ means nothing beyond “here”, and the conclusion $\Box \exists x G(x)$ reduces to $\exists x G(x)$.

Remark 19. This is the standard semantic reading of modal collapse rather than a new observation; the condition $\forall x \forall y (r x y \rightarrow y = x)$ is stated directly as a rendering of *MC* in the literature on computational variants of the argument. What Theorem 17 adds here is that it is *derived* from Scott's axioms in the formalization, with the frame hypothesis (symmetry, and nothing else) recorded.

6 One model, two jobs

Let $W = I = \{*\}$, let r be the universal relation, and put $P(\varphi) :\Leftrightarrow \varphi(*)$ — a property is positive exactly when the sole individual has it.

Proposition 20. *In this model A1a, A1b, A2, A3 and A4 all hold, and Gödel's A5, P(NE), fails.*

Proof. A1a and A1b both read $\neg\varphi(*) \rightarrow \neg\varphi(*)$. For A2, suppose $\varphi(*)$ and $\Box \forall x (\varphi(x) \rightarrow \psi(x))$; instantiating at the unique world and individual gives $\psi(*)$. For A3 we must check $G(*)$, that is $\forall \varphi (P(\varphi) \rightarrow \varphi(*))$, which unfolds to $\forall \varphi (\varphi(*) \rightarrow \varphi(*))$. A4 is immediate, there being one world. Finally P(NE) fails: it would give NE(*), and instantiating at e , which is a Gödel-essence of $*$ by Lemma 5, yields $\exists y e(y)$. $\square$

This is the non-triviality check that Theorem 7 needs. A mechanized proof of $\perp$ from a list of hypotheses is only informative if the hypotheses have not been mis-stated into absurdity; nothing in Theorem 7 would look different if A2 had been encoded in a form nothing could satisfy. Proposition 20 rules that out and isolates A5 as the axiom that tips the system over.

The same model then settles a second question.

Proposition 21. *In the same model, with the same P , Scott's A5, $P(\text{NE}_S)$, holds. Hence Scott's system is consistent.*

Proof. We must check $\text{NE}_S(*)$, that is, that every Scott-essence of $*$ is necessarily instantiated. Let $\varphi \text{ ess}_S *$. The first conjunct gives $\varphi(*)$, and with one world and one individual that is already $\Box \exists y \varphi(y)$. $\square$

The contrast between Propositions 20 and 21 is the cleanest available statement of what Gödel's omission costs. One structure, one notion of positivity; Gödel's Axiom 5 fails in it and Scott's holds, and the only difference between them is the conjunct $\varphi(x)$. It also shows that the inconsistency of Section 3 is specific to the definition of essence rather than a defect of the surrounding axioms or of our encoding of them.

This model is degenerate: one world, so $\Box$ is trivial and modal collapse holds vacuously. It is natural to ask whether that is an accident of choosing the smallest structure — whether Scott's axioms have a model with several worlds standing in some genuine accessibility relation. Theorem 17 already answers the second half, but the first half deserves an explicit answer, since a theorem forbidding modal structure would be idle if it also forbade models outright.

Proposition 22. *Let V be any type of worlds whatsoever, let r be equality on V , take one individual, and put $P(\varphi)$ at w iff that individual has φ at w . Then all of Scott's axioms hold.*

Proof. Each world sees only itself, so $\Box_r q$ at w is just q at w , and every axiom reduces to its one-world form, verified as in Proposition 21. $\square$

So the answer to “does Scott's system have a non-degenerate model?” is: *yes in cardinality, no in modal structure*. The world set may be as large as one likes — Proposition 22 places no hypothesis on V at all — but by Theorem 17 every such model is a disjoint heap of isolated copies of Proposition 21, each with its own God-like individual, none able to see any other. The one-world model is not forced; but nothing better than a heap of one-world models is available.

7 Symmetry is necessary

Symmetry is used in exactly one place, Lemma 12, to carry $\Box \exists x G(x)$ from a merely possible world back to the actual one. That is the B schema $\Diamond \Box p \rightarrow p$. Everything downstream — Theorem 13, modal collapse, Theorem 17 — inherits the hypothesis from there. It is natural to ask whether it can be weakened.

It cannot, and the reflexivity and transitivity of $S4$ are no substitute.

Proposition 23. *There is a model of Scott's axioms over a reflexive, transitive, non-symmetric frame in which Theorem 13, Theorem 14 and Theorem 17 all fail.*

Proof. Let $W = \{0, 1\}$ with $r w v$ iff $w = 0$ or $v = 1$: world 0 sees both worlds, world 1 sees only itself. This is reflexive and transitive, and not symmetric, since $r 0 1$ but not $r 1 0$. Take one individual $*$, and put

$$P(\varphi)(w) :\Leftrightarrow \varphi(*) (1) \quad \text{at both worlds.}$$

So positivity is imported wholesale from world 1, where the model of Proposition 21 lives.

A1a and A1b are immediate. For A2, the box hypothesis at either world instantiates at $v = 1$, which is accessible from both, and the argument of Proposition 21 applies. A3 asks for $G(*)$ at 1, which unfolds to $\forall\varphi(\varphi(*) \rightarrow \varphi(1))$. A4 holds because $P(\varphi)$ is a constant sentence. A5 asks for $NE_S(*)$ at 1; every world accessible from 1 is 1 itself, so the demand reduces to the first conjunct of ess_S , exactly as before.

Now $G(*)$ *fails* at world 0. Being God-like at 0 would require having at 0 every property positive there, hence every property one has at 1; and $\lambda y w. (w = 1)$ is such a property, false at 0. So there is no God-like individual at 0. Since $r00$, that refutes $\Box\exists x G(x)$ at 0, which is Theorem 13. Modal collapse fails at 0 as well: $\lambda u. (u = 0)$ holds at 0 but not at the accessible world 1. And $r01$ with $1 \neq 0$ refutes Theorem 17. $\square$

Two things follow. First, symmetry is not a convenience of the usual $S5$ presentation that a sharper proof could remove; the B schema is doing irreplaceable work, and it is doing it at precisely the step where the argument moves from possible existence to actual existence. Second, and more striking:

Corollary 24. *Scott's axioms do not entail modal collapse.*

They entail it *in B and S5* — which is where the argument is always presented, so the objection stands where it is usually made. But collapse is a consequence of the axioms *together with* a symmetric frame, not of the axioms alone, and Proposition 23 exhibits an $S4$ model of the axioms in which contingency survives. The price of the argument is therefore better stated as a price of the frame it is run in: strengthen the logic enough to deliver the conclusion, and you get the collapse with it.

Remark 25. This is consistent with, and gives one explanation of, the report in [2] that the argument needs KB — symmetry and nothing more. It also locates the target for the emendations, which is the subject of Section 8.

8 The emendations

Anderson [1] and Fitting [9] each modify the argument so as to keep the conclusion and lose the collapse. Benz Müller and Fuenmayor [3] have analysed both on the computer; we follow their presentation, and in particular their rendering of Anderson's variant as Fitting gives it. This section formalizes each variant far enough to prove its Theorem 3 and to exhibit a model in which modal collapse fails.

8.1 Anderson: biconditionals in place of A1b

Anderson's complaint is about A1b, $\neg P(\varphi) \rightarrow P(\neg\varphi)$, which says that of every property one of it and its negation is positive. That is a strong claim, and — as Section 3 showed — not one the refutation of Gödel's system ever needed. Anderson drops it. Dropping it alone breaks Theorem 2, so he strengthens the two definitions to compensate:

$$\begin{aligned} G^A(x) &:= \forall\varphi (P(\varphi) \leftrightarrow \Box\varphi(x)), \\ \varphi \text{ ess}^A x &:= \forall\psi (\Box\psi(x) \leftrightarrow \Box\forall y(\varphi(y) \rightarrow \psi(y))), \end{aligned}$$

with NE^A as before but using ess^A . The axioms are A1a, A2, A4, $\text{P}(\text{NE}^A)$, and $\text{P}(\text{G}^A)$ postulated in place of A3.

The biconditional in G^A makes uniqueness especially direct.

Proposition 26 (Monotheism). *Over a reflexive frame, Anderson’s axioms make God-like individuals unique: if $\text{G}^A(x)$ and $\text{G}^A(y)$ at w , then $y = x$.*

Proof. Let $\varphi := \lambda z u. (z = x)$. Then $\Box\varphi(x)$ holds at w , so $\text{P}(\varphi)$ at w by $\text{G}^A(x)$; so $\Box\varphi(y)$ at w by $\text{G}^A(y)$; and reflexivity gives $\varphi(y)$ at w , i.e. $y = x$. $\square$

Uniqueness is *not* peculiar to Anderson, and an earlier version of this note said it was. The same argument runs in Scott’s system, from A1b alone and with no frame condition at all: if being identical to a God-like x were not positive then its negation would be, by A1b, and x — having every positive property — would fail to be self-identical. So it is positive, and any God-like y has it. This is `Godel.scott_monotheism` in the development, and Benz Müller and Scott [4] likewise record monotheism as a consequence of the Gödel/Scott setting. What Anderson’s biconditional buys is directness, not the theorem.

Lemma 27. *Over an S5 frame, God-likeness is stable: if $\text{G}^A(x)$ at w and $r w v$, then $\text{G}^A(x)$ at v .*

Proof. Let φ be given. If $\text{P}(\varphi)$ at v then, by A4 at v and symmetry, $\text{P}(\varphi)$ at w ; so $\Box\varphi(x)$ at w ; and transitivity carries that to $\Box\varphi(x)$ at v . Conversely, if $\Box\varphi(x)$ at v then symmetry and transitivity give $\Box\varphi(x)$ at w , hence $\text{P}(\varphi)$ at w , hence $\text{P}(\varphi)$ at v by A4. $\square$

Theorem 28. *Over an S5 frame, Anderson’s axioms entail $\Box\exists x \text{G}^A(x)$.*

Read the hypothesis as what *this derivation* consumes, not as what Anderson’s system requires. Anderson himself (1990, fn. 5) notes that the weaker logic B — that is, $KT B$ — suffices for Theorem 3, and Benz Müller, Weber and Woltzenlogel Paleo [5] report it automated in the weaker KB again, and further report A4 and A5 *redundant* for this emendation, derivable from the rest. That last holds under the constant- and varying-domain readings they analyse; under the *mixed* reading Anderson himself floats — actualist quantifiers only in Theorem 3 and in the definition of essence — they find A4 still redundant, A5 independent, and a countermodel to Theorem 3 itself. Domain choice is not a free parameter.

Our proof is therefore not minimal, in frame strength or in axioms; reaching KB would need a different route, since Lemma 27 uses transitivity and the necessary-existence step uses reflexivity. An earlier version of this note reported the S5 as a cost of Anderson’s variant, which was a mistake of exactly the kind Section 11 warns about.

Proof. God-likeness is an ess^A -essence of any God-like x : the forward direction of the biconditional follows from A4 and reflexivity, the backward direction from Lemma 27. The rest is as in Theorem 13, with $\text{P}(\text{NE}^A)$ supplying necessary existence and reflexivity bringing $\Box\text{NE}^A(g)$ down to $\text{NE}^A(g)$. $\square$

8.2 Fitting: extensions in place of intensions

Fitting’s move is different and, to our eye, the more illuminating of the two. In Gödel’s and Scott’s systems P is a predicate on *intensional* properties $I \rightarrow W \rightarrow \text{Prop}$. Fitting makes it

a predicate on *extensions* — plain, world-independent sets $S : I \rightarrow \text{Prop}$ — and re-reads the definitions accordingly:

$$G^F(x) := \forall S : I \rightarrow \text{Prop}, P(S) \rightarrow Sx,$$

with essence and necessary existence likewise quantifying over extensions.

This is exactly what disarms the collapse. The proof of Theorem 14 took an arbitrary sentence p and smuggled it into the essence clause as the constant *intensional* property $\psi_p = \lambda y w. p(w)$, which tracks p from world to world. An extension has no world argument. There is no extension that varies with p , the substitution has no counterpart, and the derivation stops.

Theorem 29. *Fitting's axioms entail $\Box \exists x G^F(x)$, over an arbitrary frame — the argument runs in K .*

Proof. Theorem 1 goes through as before, giving a world v accessible from w and an individual g , existing at v , with $G^F(g)$ as evaluated at w . God-likeness is an essence of g , by the argument of Theorem 10 with extensions in place of properties; A5 then gives $\Box \exists z G^F(z)$ at w . $\square$

The absence of any frame condition here is not an accident, and Proposition 23 explains it. Symmetry was needed in Scott's system to carry a conclusion from a merely possible world back to the actual one. In Fitting's, the predicate $G^F(\cdot)$ evaluated at w is determined by the positivity facts *at* w ; a witness found at an accessible world is a witness simpliciter, and there is nothing to carry back. Freezing the predicate is what removes the need for the B schema.

8.3 Hájek: one axiom in place of two

Hájek's route is different again. He merges A1 and A2 into a single weaker axiom,

$$\text{H:A12} \quad (P(\varphi) \wedge \Box \forall x (\varphi(x) \rightarrow \psi(x))) \rightarrow \neg P(\neg \psi),$$

of which Anderson's A1a is the case $\psi := \varphi$, and which does *not* assert that ψ is positive — only that $\neg \psi$ is not. He then rebuilds God-likeness on the derived notion

$$P^\#(\varphi) := \exists \psi (P(\psi) \wedge \Box \forall y (\psi(y) \rightarrow \varphi(y))), \quad G^H(x) := \forall \varphi (\Box \varphi(x) \leftrightarrow P^\#(\varphi)),$$

keeping Anderson's essence and necessary existence unchanged. The axioms are H:A12, A3, A4 and A5.

The striking feature is that the last two are not needed.

Theorem 30. *H:A12 and A3 alone, over a symmetric frame, entail $\Box \exists x G^H(x)$. Neither A4 nor A5 is used.*

Proof. Theorem 1 goes through from H:A12: if φ were necessarily empty its entailment hypothesis would hold vacuously for every ψ , and $\psi := \neg \varphi$ makes the conclusion say that φ is not positive. Applied to A3 this gives a world v accessible from w with some g satisfying $G^H(g)$ there.

Now the key step. G^H is positive by A3, so it is necessarily implied by a positive property — itself — and hence $P^\#(G^H)$ holds outright. The biconditional in $G^H(g)$ converts that directly into $\Box G^H(g)$: the God-like individual is necessarily God-like. So $\Box \exists x G^H(x)$ holds at v , symmetry brings $\exists x G^H(x)$ back to w , and the same step applied at w finishes. $\square$

Where Scott needs A5 and an essence argument to make God-like existence necessary, Hájek’s definition delivers it outright. Benz Müller, Weber and Woltzenlogel Paleo report exactly this superfluosness, and add that A4 and A5 are nonetheless *independent* of the rest — superfluous without being redundant [5]. We have formalized the superfluosness; the independence needs a further pair of models and is not done here.

8.4 All three avoid modal collapse

Proposition 31. *Each of the three variants has a two-world model over the universal relation — an S5 frame — satisfying all of its axioms, in which modal collapse fails.*

Proof. For Fitting, take one individual existing at both worlds and call a set positive exactly when the individual belongs to it. For Anderson and for Hájek, take one individual and call a property positive exactly when the individual has it at *every* world. In each case the axioms are routine to verify, and $\lambda u. (u = w_0)$ holds at w_0 but not at the other world. $\square$

Each model also certifies the consistency of the variant it belongs to. The mechanisms differ: Fitting’s essence clause cannot express a contingent sentence at all, whereas Anderson’s and Hájek’s God-likeness are stated entirely in terms of $\Box$, so a contingent sentence never reaches their biconditionals.

Remark 32. One small thing the formalization turned up. In Scott’s system the step “what a God-like individual has is positive” (Lemma 9) is classical, because it argues by contradiction through A1b. Anderson has no A1b, and his corresponding steps — Proposition 26, Lemma 27, the essence theorem and the necessary-existence step — are all *constructive*; Lean reports them as depending on no axioms at all. Only Theorem 1, and hence Theorem 28, remains classical. Fitting keeps A1b and with it the classical step.

8.5 What is simplified here

Fitting’s own setting is an intensional higher-order modal logic with actualist quantifiers, and it supports distinctions this note does not draw. Two departures should be recorded.

First, *actualist quantifiers*. Our Fitting formalization carries an explicit existence predicate and quantifies only over individuals existing at the world in question. This is not optional: because a Fitting extension is world-independent, with constant domains “S is empty” would be a world-independent statement, Theorem 1 would yield actual rather than merely possible exemplification, and Theorem 29 would follow without ever using A5 — which is not Fitting’s argument. Our Anderson formalization, by contrast, uses constant domains, and there this costs nothing structural: his God-likeness is intensional, so $\exists x G^A(x)$ still varies from world to world and A5 still does real work.

Second, Fitting’s logic has intension and extension operators which we have not embedded, and which are needed to write his *syntax*. As Section 8.6 shows, the distinction those operators are for is nonetheless available here, because in a shallow embedding it amounts to where a world variable is bound.

8.6 De re and de dicto

The predicate $G^F(z)$ carries two worlds' worth of information: the individual z , and the world at which positivity is read off. So a claim about God-like existence has two readings,

$$\underbrace{\forall v (r w v \rightarrow \exists z (E z v \wedge G^F(z) \text{ at } w))}_{\text{de re}} \quad \underbrace{\forall v (r w v \rightarrow \exists z (E z v \wedge G^F(z) \text{ at } v))}_{\text{de dicto}}$$

and no extra machinery is needed to state them: the difference is whether the last world argument is w or v . Theorem 29 is the de re reading.

Theorem 33. *Fitting's axioms entail the de dicto reading as well, again over an arbitrary frame.*

Proof. It suffices that God-likeness is stable: if $G^F(x)$ at w and $r w v$, then $G^F(x)$ at v . By A4 the extension $\mathbf{g}_w := \{z : G^F(z) \text{ at } w\}$, positive at w by T2, is positive at v ; so anything God-like at v lies in $\mathbf{g}_w$, that is, is God-like at w . Theorem 1 supplies some z God-like at v ; that z is then God-like at w , and God-like individuals are unique at a world — by the argument of Proposition 26, which works verbatim for Fitting's system — so $z = x$ and x is God-like at v . $\square$

So with the full axiom set both readings hold. They do come apart, but earlier: on the Part I axioms A1a, A1b, A2 and T2, before A4 is available. This is the separation Benzmlüller and Fuenmayor report, and it is worth being precise about its location, since it is easy to mistake for a limitation on the conclusion.

Proposition 34. *There is a model of the Part I axioms in which God-like existence is de re possible but not de dicto possible.*

Proof. Two worlds $\{0, 1\}$ and two individuals $\{0, 1\}$. Each world sees only the other, $r w v \iff w \neq v$; the individual existing at a world is the other one, $E z v \iff z \neq v$; and a set is positive at w exactly when it contains w 's own individual.

Then $G^F(z)$ at w holds iff $z = w$: positivity at w is the principal ultrafilter at w , so the only member of every positive set is w itself. A1a, A1b and T2 are immediate. For A2, the world $\neg w$ is accessible from w and w exists there, so the entailment hypothesis instantiates to $S w \rightarrow T w$.

Now G^F is de re possible at w : the accessible world $\neg w$ holds the individual w , which exists there and is God-like as read off at w . But it is not de dicto possible, anywhere: the individual God-like at a world is that world's own, and that is precisely the one that does not exist there. $\square$

A4 must fail in this model, and does — the singleton $\{0\}$ is positive at world 0 and not at world 1. Given Theorem 33 it could not be otherwise.

Two features of the separation are worth drawing out. It needs a *non-universal* frame: A4 propagates positivity along r , so when every world sees every world it makes positivity world-independent, God-likeness world-independent with it, and the two readings identical. And it needs *varying domains*: the room for them to come apart is an individual that is God-like as read off at w but does not exist at v . That is a second reason, beyond the one in Section 8.5, why the existence predicate in this development is not optional.

9 Frame conditions, tabulated

Keeping r unconstrained in the semantics makes the following visible. Each row lists the frame condition the result actually consumes, which in two cases is weaker than the logic in which the result is usually stated.

Result	Axioms used	Frame condition
Theorem 1 (Lemma 4)	A1a, A2	none
Empty Essence (Lemma 5)	none	none
Inconsistency (Thm. 7)	A1a, A2, P(NE)	none (K)
Theorem 2 (Thm. 10)	A1b, A4	none
Theorem 3 (Thm. 13)	all of Scott's	symmetry (KB)
Modal collapse (Thm. 14)	all of Scott's	symmetry (KB)
r is equality (Thm. 17)	all of Scott's	symmetry (KB)
One world (Thm. 16)	all of Scott's	universal (S5U)
Anderson's Thm. 3 (Thm. 28)	all of Anderson's [†]	symmetry (KB) [†]
Fitting's Thm. 3 (Thm. 29)	all of Fitting's	none (K)
Hájek's Thm. 3 (Thm. 30)	H:A12, A3 <i>only</i>	symmetry (KB)

[†] These are the published values, not ours. Our Lean derivation of Anderson's Theorem 3 uses reflexivity, transitivity and symmetry and the whole axiom set; Anderson (fn. 5) and [5] give KB , with A4 and A5 redundant. Every other entry is what our proof consumes.

By Proposition 23 the symmetry entries for Scott are sharp: they fail over a reflexive transitive frame, so no weakening in the $S4$ direction is available there. The last three rows are the point of Section 8: Fitting pays no frame strength at all, Hájek pays the same as Scott but with two of his four axioms unused, and Anderson — on the published analysis — pays no more than Scott either. All three avoid the collapse, at differing frame strengths. Frame strength is not the variable that matters; the shape of the essence clause is.

Two comments. First, the refutation of Gödel's axioms needs no frame condition at all, so it is not an artefact of choosing $S5$; the axioms are inconsistent in the weakest normal modal logic. Second, the positive argument needs only symmetry, not the reflexivity and transitivity of $S5$ — which accords with the report in [2] that the argument goes through in KB . Theorem 7 is the one place where the accounting is subtle: the refutation that Leo-II found automatically does use symmetry, and so lives in KB ; the human-constructed refutation of [2] formalized here does not.

10 The Lean 4 development

The development is 1461 lines across seven files, of which 378 are code and the rest documentation, with **no dependency on Mathlib or any other library** — nothing beyond Lean's core logic is needed — and builds in a couple of seconds. The shallow embedding is direct:

```
abbrev Sentence (W : Type u) : Type u := W -> Prop
abbrev Property (I : Type v) (W : Type u) := I -> W -> Prop

def box (r : W -> W -> Prop) (p : Sentence W) : Sentence W :=
  fun w => forall v, r w v -> p v
```

Each axiom is a named predicate (`AxiomA1`, `AxiomA1b`, ...), so a theorem's hypotheses state exactly which axioms it consumes; the two axiom sets are bundled as `Godel1970 r P` and `Scott1987 r P`. The two headline statements are

```
theorem inconsistent_of_A1_A2_A5 [Nonempty W]
  (hA1 : AxiomA1 P) (hA2 : AxiomA2 r P) (hA5 : AxiomA5 r P) : False

theorem modal_collapse (hs : Symm r) (h : Scott1987 r P)
  (p : Sentence W) (w : W) (hp : p w) : box r p w
```

The first takes no hypothesis on r at all, and the second takes exactly one: that is how the middle column of Section 9 is recorded, and it is checked rather than asserted. The correspondence with this note is

<code>possibly_exists</code>	Lemma 4
<code>emptyProperty_ess</code>	Lemma 5
<code>godel_1970_inconsistent</code>	Theorem 7
<code>empty_not_scottEss</code>	Proposition 8
<code>positive_of_god</code>	Lemma 9
<code>scott_T2</code>	Theorem 10
<code>box_exists_god_of_god</code>	Lemma 11
<code>exists_god</code>	Lemma 12
<code>scott_T3</code>	Theorem 13
<code>modal_collapse</code>	Theorem 14
<code>s5_unique_world</code>	Theorem 16
<code>accessibility_eq</code>	Theorem 17
<code>box_iff, dia_iff</code>	Corollary 18
<code>toy_satisfies_A1_A2_A3_A4, toy_not_A5</code>	Proposition 20
<code>toy_satisfies_scott</code>	Proposition 21
<code>iso_satisfies_scott</code>	Proposition 22
<code>ns_satisfies_scott, ns_not_T3,</code>	
<code>ns_not_modal_collapse</code>	Proposition 23
<code>Anderson.monotheism</code>	Proposition 26
<code>Anderson.god_transfers</code>	Lemma 27
<code>Anderson.T3</code>	Theorem 28
<code>Fitting.T3</code>	Theorem 29
<code>Hajek.T3</code>	Theorem 30
<code>Anderson.noCollapse, Fitting.noCollapse,</code>	
<code>Hajek.noCollapse</code>	Proposition 31

Lean's `#print axioms` reports what each theorem ultimately rests on. Every classical result in the development reports exactly `[propext, Classical.choice, Quot.sound]`, and the following report *no axioms at all*:

<code>emptyProperty_ess</code>	does not depend on any axioms
<code>empty_not_scottEss</code>	does not depend on any axioms
<code>toy_satisfies_A1_A2_A3_A4</code>	does not depend on any axioms
<code>toy_not_A5</code>	does not depend on any axioms
<code>toy_satisfies_scott</code>	does not depend on any axioms

Three things should be read off this. First, neither Gödel's nor Scott's axioms appear anywhere, because they are not axioms of the development: they are hypotheses, discharged

by whoever claims to have a model. Every theorem is a conditional, which is what claims of this kind ought to be. Had the axioms been declared with Lean’s `axiom` command, the development would have been globally inconsistent and the certificate worthless.

Second, the only axioms that do appear are Lean’s own three — propositional extensionality, choice, and soundness of quotients — entering through the two classical steps, Lemma 4 and Lemma 9.

Third, and most usefully, the consistency of Scott’s system (`toy_satisfies_scott`) is certified constructively and axiom-free. It is an explicit model, not the output of a model finder, and can be inspected.

11 Scope of the result

The comprehension assumption. This is the one place where the results are relative to a reading. The quantifiers $\forall\varphi$ in the definitions of essence and of **NE** range over *all* properties $I \rightarrow W \rightarrow \text{Prop}$, which is what makes the empty property available for instantiation in Theorem 7 — and, on the other side, what makes the constant property ψ_p available in Theorem 14. This is the standard second-order reading. A defender could restrict the range of property quantification so as to exclude necessarily uninstantiated properties, blocking Theorem 7; but Gödel states no such restriction, and any restriction strong enough to exclude e would need independent motivation, on pain of being introduced for no reason except to save the argument. Scott’s emendation blocks the derivation without any such manoeuvre, which is a point in its favour — though note that a comprehension restriction would be no help against modal collapse, which needs only a constant property.

What is not shown. Nothing here bears on whether God exists. Theorem 7 says that one particular list of premises is self-contradictory and therefore proves everything, its intended conclusion included; an inconsistent argument is not a disproof of its conclusion, it is the absence of an argument. Theorem 13 says that another list of premises entails $\Box\exists x G(x)$; whether to accept those premises is a question about the premises, and Theorem 14 is the main reason most readers do not.

Open. Hájek’s emendations are untouched, and Fitting’s intension and extension operators are not embedded — though Section 8.6 shows that the distinction they exist to express is available without them. What a proper embedding would add is the ability to write Fitting’s own syntax, and with it the remaining readings he distinguishes, rather than the two that matter for the conclusion.

Availability

The Lean sources, the build script and the axiom certificate are at <https://github.com/alozanoroble/godel-ontological-lean>. The development builds against Lean 4 v4.34.0 with no external dependencies.

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